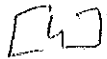


SEAT No. _____

No. of Printed Pages: 02



SARDAR PATEL UNIVERSITY
F.Y.BCA (First Semester) (CBCS) Examination 2021

Friday, 29th January, 2021

10.00 a.m. to 12.00 p.m.

US01EBCA01: Digital Computer Electronics

Maximum Marks: 70

Note: - Answers of all the questions (including multiple choice questions) should be written in the provided answer book only.

Q-1 Multiple Choice Questions.

[10]

- 1) The _____ gate has two or more input signals. All inputs must be high to get a high output.
A. AND B. OR C. NAND D. NOR.
- 2) An invert gate is also called a _____ gate.
A. NOT B. NOR C. XNOR D. NAND.
- 3) The OR gate has two or more input signals. If any input is _____, the output is high.
A. High B. Low C. Both A and B D. None
- 4) In k-map, pair eliminates _____ variable.
A. 1 B. 2 C. 3 D. 4.
- 5) The relationship between a function and its binary variables can be represented in _____.
A. Truth Table B. Decoder C. Encoder D. None
- 6) A combinational circuit that performs the arithmetic addition of two bits is called _____.
A. Full Adder B. Half Adder C. Binary Adder D. None.
- 7) A gate is a logic circuit with one or more input signals but only _____ output signal.
A. 2 B. 1 C. 3 D. 4.
- 8) In half adder XOR gate's output is _____.
A. Carry B. Sum C. Reminder D. None.
- 9) In shift left register, the arrival of the first rising clock edge sets the _____ flip-flop.
A. left B. right C. up D. down.
- 10) A multiplexer also called a _____.
A. Data Multiplier B. Data Selector C. Inverter D. Remover

Q-2 Fill in the Blanks and True-False

[8]

1. An invert gate is also called a _____ gate.
2. The NAND gate has two or more input signals. If all inputs are _____, the output is low.
3. A _____ is logic circuit that can add two binary numbers.
4. The full CKT adds _____ digit at a time.
5. In D flip-flop, when CLK is high then output is same as output. (True/False)
6. A buffer register is the simplest kind of register; all it does store a digital

(1)

(P.T.O)

word.(True/False)

7. A multiplexer also called a Data Inverter.(True/False)
8. in half adder and gate's output is carry.(True/False)

Q-3 Answer the following questions (Any Ten).

[20]

- 1 Draw circuit for **AND, NOR** gate.
- 2 Draw the circuit for $A'B+B'C+AC'$
- 3 Describe **Pair** in K-Map
- 4 Simplify using K-Map $F(A,B,C)=\Sigma(1,2,5)$
- 5 Describe **Half adder** in short.
- 6 Describe **Binary Adder** in short.
- 7 Draw Circuit of **Shift Left Register**.
- 8 Draw Circuit of **Controlled buffer register**.
- 9 Draw Circuit of **Ring Counter**.
- 10 Define **Decoder** in short.
- 11 Define **Encoder** in short.
- 12 Explain **De Morgan's first theorem**.

Q-4 Long Answer Questions(Attempt any 4 out of 8).

[32]

1. Simplify this Boolean expression and draw circuit: $ABC'+ABC+A'BC$
2. Explain **AND, OR, NOT** and **XOR**.
3. Simplify this using k-map $F(A,B,C,D)=\Sigma(1,3,5,6,8,11,15)$
4. Explain **3x8 Line Decoder** in detail.
5. Explain **Binary Adder-Subtractor** in detail.
6. Explain **4x1 Multiplexer** in detail.
7. Explain **D flip-flop** in detail.
8. Explain **Controlled Buffer Register**.

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